

gui.property\_panel.py

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from PySide6.QtWidgets import QFrame, QVBoxLayout, QGroupBox, QFormLayout,
QComboBox, QDoubleSpinBox, QLineEdit, QLabel
from PySide6.QtCore import Qt
from PySide6.QtGui import QPixmap
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class PropertyPanel(QFrame):
    def __init__(self, database_manager):
        super().__init__()
        self.database_manager = database_manager
        self.setFrameStyle(QFrame.StyledPanel | QFrame.Sunken)
        self.setFixedWidth(300)
        self.main_layout = QVBoxLayout(self)

        self.setup_fluid_section()
        self.setup_element_section()
        self.setup_common_info()
        self.setup_tank_properties()
        self.setup_pipe_properties()
        self.main_layout.addStretch()

    def setup_fluid_section(self):
        group = QGroupBox("Configuración del Fluido")
        form = QFormLayout(group)
        self.fluid_combo = QComboBox()
        form.addRow("Fluido Global:", self.fluid_combo)
        self.main_layout.addWidget(group)

    def setup_element_section(self):
        self.element_group = QGroupBox("Propiedades del Elemento")
        form = QFormLayout(self.element_group)

        self.id_label = QLabel("-")
        self.tag_input = QLineEdit()
        self.material_combo = QComboBox()
        self.schedule_combo = QComboBox()
        self.size_combo = QComboBox()
        self.length_spin = QDoubleSpinBox()
        self.length_spin.setSuffix(" m")

        form.addRow("ID Interno:", self.id_label)
        form.addRow("Etiqueta:", self.tag_input)
        form.addRow("Material:", self.material_combo)
        form.addRow("Cédula (Schedule):", self.schedule_combo)
        form.addRow("Diámetro Nom:", self.size_combo)
        form.addRow("Longitud:", self.length_spin)

        self.main_layout.addWidget(self.element_group)

    def setup_common_info(self):
        group = QGroupBox("Información General")
        form = QFormLayout(group)
        self.id_label = QLabel("-")
        self.tag_input = QLineEdit()
        form.addRow("ID Interno:", self.id_label)
        form.addRow("Etiqueta:", self.tag_input)
        self.main_layout.addWidget(group)

    def setup_tank_properties(self):
        """Crea la interfaz específica para tanques con tu imagen"""
        self.tank_group = QGroupBox("Propiedades del Tanque")
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        layout = QVBoxLayout(self.tank_group)

        # Imagen de referencia (Tu dibujo)
        self.tank_img_label = QLabel()
        pix = QPixmap("resources/images/tank_diagram.png").scaledToWidth(250,
Qt.SmoothTransformation)
        self.tank_img_label.setPixmap(pix)
        self.tank_img_label.setAlignment(Qt.AlignCenter)
        layout.addWidget(self.tank_img_label)

        # Campos técnicos de PipeFlow
        form = QFormLayout()
        self.liquid_level = QDoubleSpinBox()
        self.liquid_level.setSuffix(" m")
        self.surf_press = QDoubleSpinBox()
        self.surf_press.setSuffix(" bar")

        form.addRow("Nivel Líquido:", self.liquid_level)
        form.addRow("Presión Superficie:", self.surf_press)
        layout.addLayout(form)

        self.main_layout.addWidget(self.tank_group)
        self.tank_group.hide() # Oculto por defecto

def setup_pipe_properties(self):
    """Mantiene los campos de tubería que ya tenías"""
    self.pipe_group = QGroupBox("Propiedades de Tubería")
    form = QFormLayout(self.pipe_group)
    # ... (material_combo, size_combo, length_spin, etc.)
    self.main_layout.addWidget(self.pipe_group)
    self.pipe_group.hide()

# --- LOGIC METHODS (ENGLISH) ---
def populate_initial_data(self):
    # Fluids from 'liquids' table
    for f_id, name, temp in self.database_manager.get_fluids():
        self.fluid_combo.addItem(f"{name} ({temp}°C)", f_id)

    # Materials from 'pipes_mm'
    for mat in self.database_manager.get_distinct_materials():
        self.material_combo.addItem(mat)

    self.material_combo.currentIndexChanged.connect(self.update_schedules)
    self.schedule_combo.currentIndexChanged.connect(self.update_sizes)
    self.update_schedules()

def update_schedules(self):
    self.schedule_combo.blockSignals(True)
    self.schedule_combo.clear()
    material = self.material_combo.currentText()
    schedules = self.database_manager.get_schedules_by_material(material)
    self.schedule_combo.addItems(schedules)
    self.schedule_combo.blockSignals(False)
    self.update_sizes()

def update_sizes(self):
    self.size_combo.clear()
    material = self.material_combo.currentText()
    schedule = self.schedule_combo.currentText()
    sizes = self.database_manager.get_sizes(material, schedule)
    for rowid, nominal in sizes:
        self.size_combo.addItem(str(nominal), rowid)

def update_node_data(self, node_data):

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self.id_label.setText(str(node_data.id))
self.tag_input.setText(node_data.notes)

def update_node_data(self, node_dict):
    """Cambia dinámicamente la visibilidad según el tipo de nodo"""
    node_type = node_dict.get("type", "Junction")
    data = node_dict["data"]

    self.id_label.setText(str(data.id))
    self.tag_input.setText(data.notes)

    if node_type == "Tank":
        self.tank_group.show()
        self.pipe_group.hide()
    elif node_type == "Junction":
        self.tank_group.hide()
        self.pipe_group.hide()
    else:
        # Para tuberías u otros elementos
        self.tank_group.hide()
        self.pipe_group.show()

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